

Real-Time Evaluation and Few-Shot Hardware Adaptation of a Vision-Based Tactile Sensor

Juhee Park, Seojung Min, and Jung Kim*

Department of Mechanical Engineering, KAIST,
Daejeon, 34141, Korea

(annaqkr@kaist.ac.kr; seojung.min@kaist.ac.kr; jungkim@kaist.ac.kr) * Corresponding author

Abstract: Vision-based tactile sensing is increasingly judged by offline test-set accuracy, yet the quantity that matters for robotics is accuracy under live contact on the deployed hardware. Sensors such as DIGIT reach high accuracy on offline texture-classification benchmarks, but this accuracy can fall under real-time robotic contact and degrade further when the gel pad or sensor unit is replaced. We quantify these effects on a DIGIT sensor mounted on a 7-DOF Kinova Gen2 arm classifying five sandpaper grit levels (P100–P400) under real-time robotic contact. A model that reaches 96.76% on a static test set drops to 82.96% in real time on the same hardware, and falls to 72.22% on a previously unseen sensor-gel configuration. Fine-tuning on as few as 14–28 images per class recovers the gap, reaching a peak of 88.15% at 70 images per class. Offline accuracy is therefore an unreliable indicator of deployed performance, but the cost of adapting to a new hardware configuration can be small.

Keywords: Vision-based tactile sensor, texture classification, real-time evaluation, hardware adaptation, few-shot fine-tuning.

1. INTRODUCTION

Vision-based tactile sensors such as GelSight [1] and DIGIT [2] place a camera behind a soft elastomer gel and record the gel’s deformation upon contact as a high-resolution RGB image. Because surface microstructure is rendered directly into pixels, these sensors are attractive for fine-grained perception tasks such as texture and material recognition, where the differences between classes are too small to resolve with force or proprioceptive signals alone. A growing body of work reports strong classification accuracy on tactile image datasets, and the field increasingly relies on offline test-set accuracy as the headline metric.

For a deployed robot, however, what matters is accuracy under live contact rather than on a static test split, and two practical issues appear only there. First, a model that performs well on a held-out test set can degrade when run in real time on a robot, because live contact involves transient dynamics, vibration, and pressure fluctuations that are absent from a static dataset. Second, the captured image is a rendering of a physical process that depends on the specific elastomer gel in front of the camera. When the gel pad is replaced or a different sensor unit is used, the image distribution shifts and the learned classifier no longer applies cleanly. Sensor-to-sensor and gel-related variability is widely acknowledged as a source of poor generalization for data-driven tactile models [3], [4], [5], [6], [7], and recent work has begun to address it through cross-sensor representations [6] and few-sample adaptation across sensors [4], [5], [7]. The offline-to-deployment accuracy gap, in contrast, has received less direct attention in vision-based tactile sensing, and how much target-domain data is needed to close both gaps under real-time robot evaluation has not been systematically reported.

This paper measures both effects on a five-class sandpaper grit classification task (P100–P400) using a DIGIT

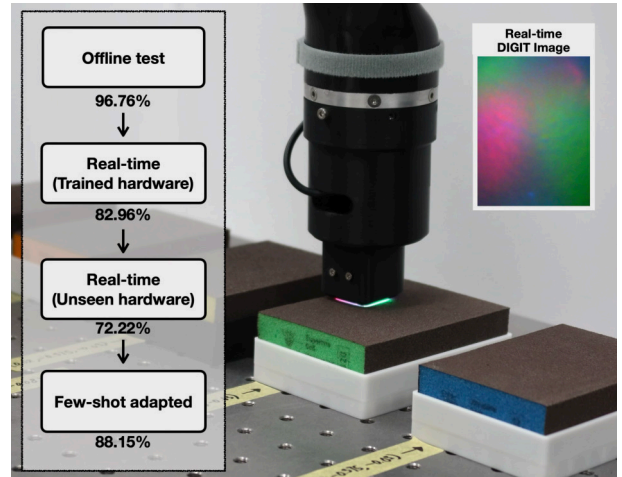


Fig. 1. Real-time five-class sandpaper grit classification with a DIGIT sensor on a Kinova Gen2 arm. The overlay traces accuracy across offline test, real-time deployment, unseen hardware, and few-shot adaptation, showing that offline accuracy overstates deployed performance while adaptation is inexpensive.

sensor mounted on a 7-DOF Kinova Gen2 arm (Fig. 1). Sandpaper grit provides a controlled, monotonic axis of texture difficulty. All evaluation is performed in real time with the sensor pressing physical samples, not only on a static test split.

The contributions of this paper are:

- A real-time, on-robot characterization of the offline-to-deployment accuracy gap for a vision-based tactile classifier.
- A quantification of the additional drop introduced by an unseen sensor-gel configuration without target-domain adaptation.
- An adaptation-data-requirement curve characterizing how few target-domain images suffice to recover deployed accuracy.

2. RELATED WORK

Tactile texture and material classification. Texture, material, and surface-property recognition from tactile signals has been studied with a range of sensors and learning approaches. Yuan et al. [8] used GelSight images to estimate hardness in a shape-independent manner. Gao et al. [9] combined visual and haptic data for material understanding. ViTac [10] shared features between vision and tactile inputs for cloth texture recognition, and Pastor et al. [11] applied 3D CNNs to tactile object recognition under robotic palpation. More recently, Yan et al. [12] addressed surface texture recognition with vision-based tactile sensing. These approaches typically report accuracy on offline test sets collected on a fixed sensor unit; how the reported numbers transfer to live robotic contact, or to a different sensor unit or gel pad, is less often addressed.

Cross-sensor and cross-gel generalization. Hardware variability is widely recognized as a source of poor generalization for vision-based tactile models [3], and several recent works address it. Sparsh [6] learns self-supervised touch representations on a large unlabeled corpus spanning multiple vision-based tactile sensors, aiming for sensor-agnostic features. FeelAnyForce [5] predicts contact forces across GelSight Mini and DIGIT units using a calibration procedure for adapting a pre-trained model. FEATS [7] estimates force distributions on GelSight Mini and reports that errors increase on an unseen sensor with a new gel, and that adding gel-swapped training samples mitigates this. Jing and Qian [4] reduce cross-sensor domain gaps via few-sample style-to-content unsupervised domain adaptation, reporting strong recognition accuracy when transferring to a DIGIT sensor.

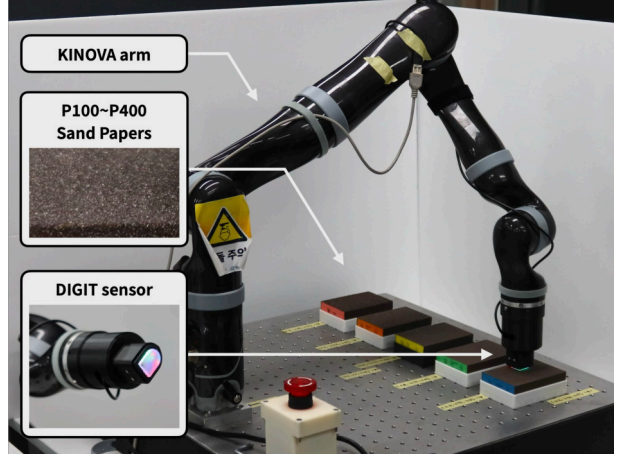
Position of this work. Together, these two lines of work cover (i) what tactile sensors can classify on stored data, and (ii) how learned representations behave across sensors and gels. We focus on a complementary question that sits between them. For a fixed sensor family (DIGIT) and a texture classification task, we examine how accuracy changes when the model is moved from an offline test split to a live robot, and how much target-domain data is needed to recover accuracy on a previously unseen sensor-gel configuration. Our contribution is empirical rather than methodological. We use a deliberately standard pre-train-then-fine-tune recipe and report what the data say.

3. EXPERIMENTAL METHODOLOGY

3.1 Sensor and Platform

We use the DIGIT vision-based tactile sensor [2], which streams VGA (640×480) RGB images of gel deformation at 30 fps. The sensor is mounted on the end-effector of a 7-DOF Kinova Gen2 arm (Fig. 2(a)). The arm controls contact location and indentation depth, and holds contact while frames are captured. Across our ex-

(a) Hardware platform



(b) Real-time classification

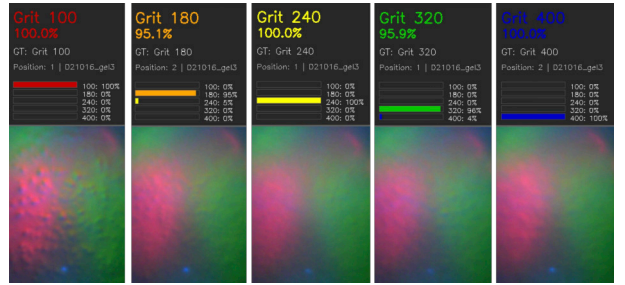


Fig. 2. (a) Hardware platform. DIGIT sensor on a Kinova Gen2 arm pressing one of five sandpaper grades (P100–P400). (b) Representative real-time classification snapshots for each grit class.

periments we use three DIGIT units (D21016, D21076, D21345) and four elastomer gel pads (gel1–gel4), allowing controlled variation of the (sensor unit, gel pad) configuration.

3.2 Task and Sandpaper Samples

We classify five sandpaper grit levels, P100, P180, P240, P320, and P400 (Fig. 2(a)). Grit number indexes texture fineness monotonically, which makes the task a controlled difficulty ladder. The coarsest grade (P100) is easily separable, while the finest grades (P320, P400) approach the spatial resolution that DIGIT can resolve. The samples are mounted on an optical breadboard with fixed positions known to the robot. A separate *no-contact* class is also collected so the live system can reject frames in which the sensor is not actively pressing a sample.

3.3 Data Collection

Data are collected from two complementary sources to balance volume and diversity. *Robotic collection* uses a scripted contact trajectory over a grid of positions and indentation depths; it is highly repeatable and scales easily but produces low-diversity data. *Manual collection* varies contact angle, force, and orientation that the scripted protocol does not cover. The dataset is constructed to span multiple sensor-gel configurations rather than a single one, on the hypothesis that exposing the classifier to several hardware states at training time re-

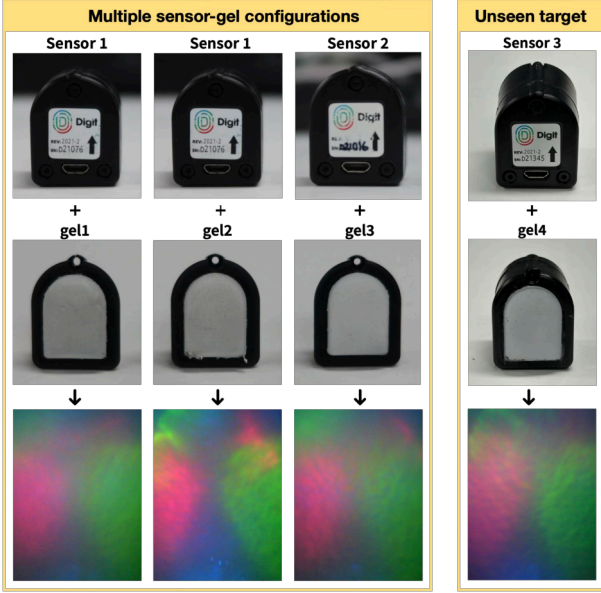


Fig. 3. The four sensor-gel configurations used in this paper. Three (D21076+gel1, D21076+gel2, D21016+gel3) are included in training; D21345+gel4 is held out as the unseen target.

duces its dependence on any one of them and lowers the cost of adapting to a new configuration at deployment. In total, seven recording sessions across two DIGIT units (D21016, D21076) and three gel pads (gel1, gel2, gel3) yield a balanced five-class dataset of 34,546 images split 70%/15%/15% into train (24,182), validation (5,182), and test (5,182). A separate held-out configuration (D21345 with a previously unused gel pad, gel4) is reserved as the unseen target in Section 4 (Fig. 3).

3.4 Real-Time Evaluation Protocol

Static test-set accuracy on stored images is not a reliable predictor of deployed performance, because live operation introduces transient contact dynamics, vibration, and pressure fluctuations. We therefore evaluate the model on the actual robot following a fixed contact protocol. For each grit class, the sensor is pressed at a 2×3 grid of positions on the sandpaper sample. At each position three press-depth levels (light, medium, firm) are applied, and three independent contact samples are recorded per depth level, giving $2 \times 3 \times 3 \times 3 = 54$ real-time trials per class and 270 trials per sensor-gel configuration. Fig. 2(b) shows representative real-time classification snapshots, including the predicted class, confidence, and ground-truth label. A trial is counted correct when the predicted class on a single frame matches the ground truth. The same protocol is used for every configuration so that real-time accuracy numbers are directly comparable.

3.5 Model and Training

The training procedure is intentionally a standard pre-train-then-fine-tune recipe applied to the multi-configuration dataset described above. The hypothesis to be tested is whether broader hardware coverage at pre-

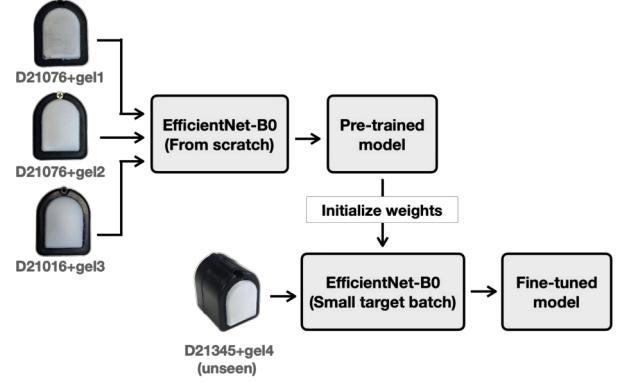


Fig. 4. Method overview. The classifier (EfficientNet-B0) is pre-trained on three sensor-gel configurations, then fine-tuned on a small target-domain batch from a previously unseen configuration.

training time, combined with a small fine-tuning batch at deployment, is sufficient to close the gaps observed in real-time evaluation.

We use EfficientNet-B0 [13] as the classifier. The model is trained from scratch on the hybrid dataset, rather than from ImageNet weights, because the domain-specific tactile images differ structurally from natural images at this data scale. Input images are resized to 224×224 and scaled to $[0, 1]$; we apply neither data augmentation nor ImageNet normalization, as aggressive augmentation degraded performance in our earlier experiments. Training uses the Adam optimizer and cross-entropy loss, with a learning rate of 1×10^{-4} , batch size 32, and 50 epochs, keeping the best-validation checkpoint. For adaptation to a new sensor-gel configuration, we fine-tune from the pre-trained weights with a ten-times-lower learning rate (1×10^{-5}), batch size 16, and up to 20 epochs with early stopping (patience 10). All data splits use a fixed random seed.

4. RESULTS

4.1 Offline Test Performance

We first report performance on the held-out static test split. The classifier reaches 96.76% overall accuracy with a mean prediction confidence of 0.9923. Per-class accuracies are 99.14% (P100), 97.89% (P180), 95.95% (P240), 92.82% (P320), and 97.96% (P400). P320 is the weakest class, with errors falling mainly into the adjacent fine grade P400 (confusion matrix in Fig. 5(a)). Taken on its own, this static result would suggest a well-calibrated, near-saturated model.

4.2 Real-Time Evaluation on Known Hardware

We next deploy the same classifier on the robot and evaluate it in real time on three sensor-gel configurations seen during training, D21076+gel1, D21076+gel2, and D21016+gel3. Real-time accuracy is reported in Table 1.

Two observations follow. First, the static-to-real-time gap is large and consistent. On the two config-

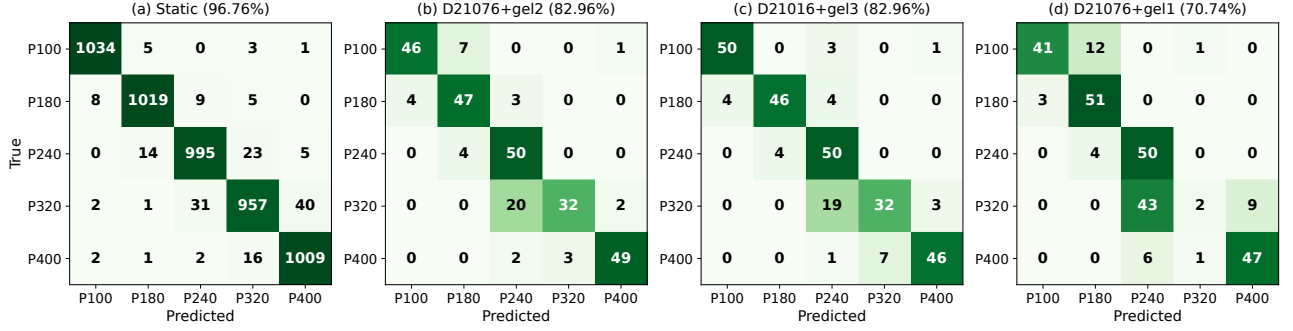


Fig. 5. Confusion matrices for the pre-trained classifier. (a) Static test set. (b–d) Real-time evaluation on three in-distribution sensor-gel configurations. The dominant P320 confusion shifts from P400 in the static setting to P240 under real-time contact.

Table 1. Real-time accuracy (%) of the pre-trained classifier by sensor-gel configuration. “Static” is the held-out test split for reference. Columns D21076+gel2 and D21016+gel3 differ in sensor unit but share a comparable gel state; D21076+gel1 shares the sensor unit (D21076) with the first real-time column but uses a different gel pad.

Class	Static	D21076+gel2	D21016+gel3	D21076+gel1
P100	99.14	85.19	92.59	75.93
P180	97.89	87.04	85.19	94.44
P240	95.95	92.59	92.59	92.59
P320	92.82	59.26	59.26	3.70
P400	97.96	90.74	85.19	87.04
Overall	96.76	82.96	82.96	70.74

urations with comparable gel state (D21076+gel2 and D21016+gel3), real-time accuracy is identically 82.96%, a drop of about 14 percentage points from the 96.76% static result. The same model does considerably worse when actually pressing a sample than when scoring stored images, even though the hardware itself is the same as during training. Live operation introduces transient contact dynamics, vibration, and pressure fluctuations that the static test split does not contain, and these effects alone account for a substantial gap. Second, the gap is not uniform across classes. P320 collapses from 92.82% to 59.26%, and the dominant confusion shifts from the adjacent fine grade P400 (its static neighbor) toward the coarser P240 (Fig. 5(b,c)). Live contact dynamics affect the fine-grit patterns differently from static pressing.

The third configuration, D21076+gel1, drops further to 70.74%. On this configuration P320 falls to 3.70%, essentially chance. Two of the three in-distribution configurations therefore converge to the same real-time performance level, while the third is substantially worse, indicating that the (sensor, gel) pair affects deployed accuracy in ways that the static evaluation does not capture. We treat 82.96% as the in-distribution real-time ceiling for the remainder of this paper, as it is the level reached

on the two comparable in-distribution configurations.

4.3 Real-Time Evaluation on an Unseen Sensor-Gel Configuration

We then evaluate the same classifier, without any adaptation, on a previously unseen sensor-gel configuration, DIGIT unit D21345 paired with gel pad gel4, neither of which appeared in training. To gauge the variability of real-time evaluation on this configuration, we ran the same 270-trial protocol three independent times, obtaining 77.04%, 72.59%, and 67.04% (mean 72.22%, range 10 percentage points). We use this mean of 72.22% as the zero-shot baseline (Fig. 6). It lies a further 10.74 percentage points below the in-distribution real-time ceiling of 82.96% and 24.54 percentage points below the offline result. The same model that scored 96.76% on stored images now classifies under three quarters of live contacts correctly when the hardware in front of the camera differs from any configuration seen during training. The offline number, taken alone, would have overstated real-time performance on this hardware by roughly a factor of $0.9676/0.7222 \approx 1.34$. The 10-percentage-point spread across the three trials is itself informative. Real-time tactile evaluation has non-trivial intrinsic variability, which should be kept in mind when reading any single-number accuracy on this kind of system.

4.4 Adaptation Data Requirements

We finally ask how much target-domain data is needed to close this gap. We fine-tune the pre-trained model on small labeled batches of images from the unseen D21345+gel4 configuration and re-measure real-time accuracy after each round. Each round adds 14 images per class (collected with the same contact protocol used elsewhere in this paper); we report up to seven rounds in Table 2 and Fig. 6.

The recovery is fast. With 14 images per class, real-time accuracy rises from 72.22% to 81.48%, recovering $9.26/(82.96 - 72.22) \approx 86\%$ of the hardware-induced gap. With 28 images per class, accuracy reaches 83.70%, surpassing the in-distribution real-time ceiling. The fine-tuning curve continues to climb gradually and peaks at 88.15% with 70 images per class. Beyond that point, accuracy declines (85.56% at 84 images per class, 82.22%

Table 2. Fine-tuning the pre-trained classifier on the unseen configuration D21345+gel4. Each round adds 14 images per class to the adaptation set.

Round	Images/class	Real-time acc. (%)	Δ (pp)
Baseline	0	72.22	—
1	14	81.48	+9.26
2	28	83.70	+2.22
3	42	87.41	+3.71
4	56	87.41	+0.00
5	70	88.15	+0.74
6	84	85.56	-2.59
7	98	82.22	-3.34

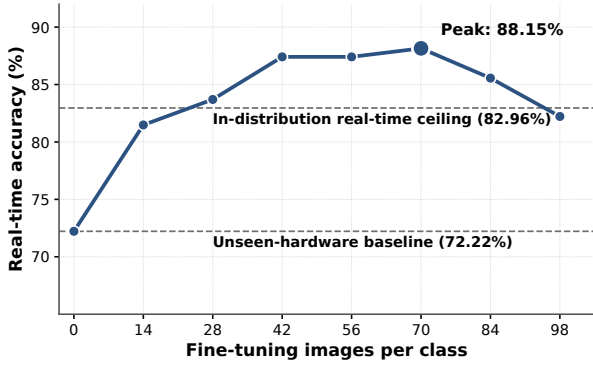


Fig. 6. Real-time accuracy on the unseen configuration D21345+gel4 as a function of per-class adaptation data. Accuracy recovers most of the gap by 14–28 images per class and peaks at 88.15% before declining.

at 98); at this scale of evaluation (54 trials per class), several factors could contribute, including statistical noise and drift in contact conditions during the extended collection, and we do not attempt to attribute the decline to a single cause. The practical takeaway is that adaptation gains saturate quickly. As few as 14 images per class are enough to recover most of the gap, and 28 images per class are enough to match the performance the same model attains on hardware it was trained on.

A visual summary of the four operating points is shown in Fig. 7.

5. DISCUSSION

Offline accuracy is an unreliable indicator of deployed performance. The clearest message from our results is that a static test split substantially overestimates what the same model achieves on the robot. Even on hardware identical to that used during training, real-time accuracy is roughly 14 percentage points lower than the static number. This suggests that benchmark scores reported on offline tactile datasets should be read with

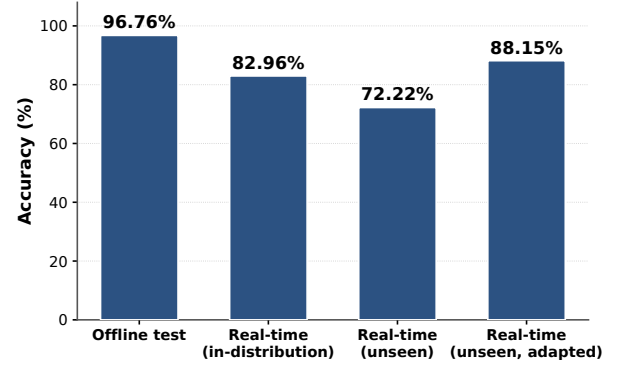


Fig. 7. Summary of the four operating points measured in this paper. Offline accuracy (96.76%) drops to 82.96% in real time and 72.22% on unseen hardware, then recovers to 88.15% with few-shot adaptation.

caution, particularly for tasks that will ultimately be deployed on a robot, and that real-time, on-robot evaluation is an informative complement rather than a redundant check.

Adapting to a new sensor-gel configuration is inexpensive. The second message is that the additional drop introduced by an unseen sensor-gel configuration—in our case 10.74 percentage points below the in-distribution real-time ceiling—is largely recoverable with very little target-domain data. A bounded collection of 14–28 images per class is enough. Re-collecting an entire dataset for every gel replacement or sensor swap is therefore not required. This has practical implications for deploying vision-based tactile classifiers in settings where hardware is replaced periodically (for example, due to gel wear) or where multiple sensor units are used.

Broader hardware coverage during pre-training appears to lower the adaptation cost. The zero-shot accuracy on the completely unseen configuration (72.22%, 87% of the in-distribution real-time ceiling) is relatively high considering that neither the sensor unit nor the gel pad appeared in training. This is consistent with the working hypothesis behind collecting data across multiple sensor-gel configurations. A classifier exposed to several hardware states at training time should be less tied to any single one, leaving a smaller residual gap to close at deployment. We did not isolate this effect with a single-configuration baseline, however, so the contribution of multi-configuration pre-training to the observed zero-shot accuracy cannot be quantified here; a controlled comparison is left to future work.

Scope and limitations. Our results are a focused case study on one texture classification task and one sensor family (DIGIT). The experimental design does not separate the effect of the sensor unit from that of the gel pad. Each (sensor, gel) pair is treated as a single configuration. We also evaluate adaptation on a single unseen configuration; whether the same data budgets generalize to other tasks and sensor families remains an open question. Expanding the hardware coverage at pre-training time—more sensor units and more gel pads—could plausibly

raise the zero-shot baseline further and reduce the per-configuration fine-tuning effort, and is a natural direction for a follow-up study.

6. CONCLUSION

We quantified, on a five-class sandpaper grit classification task with a DIGIT sensor on a Kinova Gen2 arm, how the accuracy of a vision-based tactile classifier changes between offline evaluation, real-time evaluation on the same hardware, and real-time evaluation on a previously unseen sensor-gel configuration, and how much target-domain data is needed to recover the resulting gap. Offline accuracy (96.76%) overstated real-time accuracy on in-distribution hardware (82.96%) by about 14 percentage points, and real-time accuracy fell further to 72.22% on a previously unseen sensor-gel configuration. Fine-tuning on as few as 14 images per class recovered 86% of the hardware-induced gap, 28 images per class restored the in-distribution real-time level, and 70 images per class reached a peak of 88.15%. In this setup, the results suggest that offline benchmarks alone do not reliably predict deployed performance, but that adapting a vision-based tactile classifier to a new hardware configuration can be done with very little data. We see two natural extensions: characterizing how these data budgets transfer to other vision-based tactile tasks (slip detection, hardness estimation, material recognition on non-abrasive surfaces) and to other sensor families, and isolating sensor-unit and gel-pad effects with a more controlled experimental design.

ACKNOWLEDGEMENT

This work was supported in part by the National Research Council of Science and Technology as part of the project titled “Development of core technologies for robot general purpose task artificial intelligence (Ro-GeTA) framework” under Grant NK261B.

REFERENCES

- [1] W. Yuan, S. Dong, and E. H. Adelson, “GelSight: High-resolution robot tactile sensors for estimating geometry and force,” *Sensors*, vol. 17, no. 12, 2762, 2017.
- [2] M. Lambeta, P.-W. Chou, S. Tian, B. Yang, B. Maloon, V. R. Most, D. Stroud, R. Santos, A. Byagowi, G. Kammerer, D. Jayaraman, and R. Calandra, “DIGIT: A novel design for a low-cost compact high-resolution tactile sensor with application to in-hand manipulation,” *IEEE Robotics and Automation Letters*, vol. 5, no. 3, pp. 3838–3845, 2020.
- [3] H. Li, Y. Lin, C. Lu, M. Yang, E. Psomopoulou, and N. F. Lepora, “Classification of vision-based tactile sensors: A review,” *IEEE Sensors Journal*, vol. 25, no. 19, pp. 35672–35686, 2025.
- [4] X. Jing and K. Qian, “Reducing cross-sensor domain gaps in tactile sensing via few-sample-driven style-to-content unsupervised domain adaptation,” *Sensors*, vol. 25, no. 1, 256, 2025.
- [5] A.-H. Shahidzadeh, G. Caddeo, K. Alapati, L. Natale, C. Fermüller, and Y. Aloimonos, “FeelAnyForce: Estimating contact force feedback from tactile sensation for vision-based tactile sensors,” in *Proc. of the IEEE International Conference on Robotics and Automation (ICRA)*, pp. 251–257, 2025.
- [6] C. Higuera, A. Sharma, C. K. Bodduluri, T. Fan, P. Lancaster, M. Kalakrishnan, M. Kaess, B. Boots, M. Lambeta, T. Wu, and M. Mukadam, “Sparsh: Self-supervised touch representations for vision-based tactile sensing,” in *Proc. of the 8th Conference on Robot Learning (CoRL)*, 2024.
- [7] E. Helmut, L. Dziarski, N. Funk, B. Belousov, and J. Peters, “Learning force distribution estimation for the GelSight Mini optical tactile sensor based on finite element analysis,” in *Proc. of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 8553–8560, 2025.
- [8] W. Yuan, C. Zhu, A. Owens, M. A. Srinivasan, and E. H. Adelson, “Shape-independent hardness estimation using deep learning and a GelSight tactile sensor,” in *Proc. of the IEEE International Conference on Robotics and Automation (ICRA)*, pp. 951–958, 2017.
- [9] Y. Gao, L. A. Hendricks, K. J. Kuchenbecker, and T. Darrell, “Deep learning for tactile understanding from visual and haptic data,” in *Proc. of the IEEE International Conference on Robotics and Automation (ICRA)*, pp. 536–543, 2016.
- [10] S. Luo, W. Yuan, E. Adelson, A. G. Cohn, and R. Fuentes, “ViTac: Feature sharing between vision and tactile sensing for cloth texture recognition,” in *Proc. of the IEEE International Conference on Robotics and Automation (ICRA)*, pp. 2722–2727, 2018.
- [11] F. Pastor, J. M. Gandarias, A. J. García-Cerezo, and J. M. Gómez-de-Gabriel, “Using 3D convolutional neural networks for tactile object recognition with robotic palpation,” *Sensors*, vol. 19, no. 24, 5356, 2019.
- [12] Y. Yan, Z. Hu, Y. Shen, and J. Pan, “Surface texture recognition by deep learning-enhanced tactile sensing,” *Advanced Intelligent Systems*, vol. 4, no. 1, 2100076, 2022.
- [13] M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proc. of the International Conference on Machine Learning (ICML)*, pp. 6105–6114, 2019.